

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

April – May 2016

Max. Marks:100

Class: SE

Name of the Course: Applied Mathematics – III

Course Code : UITC301/UCEC 301

Duration: 3hr.

Semester: III

Branch: IT /COMP

KJSC 2014 (AUTO)

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.	Questions	Marks
Q.1	A Attempt any THREE of the following.	18
(i)	Evaluate: $\int_0^{\infty} e^{-t} \left(\int_0^t u \cos^2 u du \right) dt$	
(ii)	Find $L \{ (1+3t-4t^2+2t^3) H(t-3) \}$	
(iii)	Find the L T of $f(t) = t, \quad 0 < t < \pi; \quad f(t) = \pi - t, \quad \pi < t < 2\pi$	
(iv)	Find $L^{-1} \left[\frac{s^2}{(s^2+4)^2} \right]$ by convolution theorem.	
	B Attempt any ONE of the following.	7
(i)	Solve $\frac{d^2 x}{dt^2} + 9x = \cos 2t, x(0) = 1, x\left(\frac{\pi}{2}\right) = -1$ by Laplace Transform.	
(ii)	Find Inverse Laplace Transform of $\tan^{-1}\left(\frac{1}{s^2}\right)$	
Q.2	A Attempt any THREE of the following.	18
(i)	Find the Fourier series for the function $f(x) = x^2$ in $(0, 2\pi)$ and hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$	
(ii)	Find the Fourier series for the function $f(x) = x \cos x$ in $(-\pi, \pi)$	
(iii)	Show that the functions $f_1(x) = 1, f_2(x) = x$ are orthogonal in $(-1, 1)$. Determine the constants a and b such that the function $f_3(x) = -1 + ax + bx^2$ is orthogonal to both f_1 and f_2 on that interval.	
(iv)	Find Complex form of Fourier series for $F(X) = \cosh x + \sinh x$ in $(-\pi, \pi)$.	

	B	Attempt any ONE of the following.	7
		(i) Express the function $f(x) = \sin x$, $0 < x \leq \pi$ $= 0$, $x < 0, x > \pi$ as Fourier integral and prove $F(x) = \frac{1}{\pi} \int_0^{\infty} \frac{\sin wx + \cos w(\pi - x)}{1 - w^2} dw$ Hence deduce that $\int_0^{\infty} \frac{\cos(w\pi/2)}{1 - w^2} dw = \frac{\pi}{2}$ (ii) Find Half Range cosine series for $f(x) = x$, $0 < x < 2$, Using Parseval's identity, deduce that i) $\frac{\pi^2}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$ ii) $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$	
Q.3	A	Attempt any THREE of the following.	18
		(i) Find eigen values & eigen vectors of $A^3 - A^2 + 4I$ and if $A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$	
		(ii) Is the following matrix A derogatory? Justify your answer. $A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}$	
		(iii) Find $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ If $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$	
		(iv) Find $\cos A$. Where $A = \begin{bmatrix} \pi & \frac{\pi}{4} \\ 0 & \frac{\pi}{2} \end{bmatrix}$	
	B	Attempt any ONE of the following.	7
		(i) Is the following matrix diagonalisable? If yes find diagonal form and transforming matrix. $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$	
		(ii) Verify the Cayley Hamilton Theorem for the matrix A and hence find A^{-1} & A^4 where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$	
Q.4	A	Attempt any THREE of the following.	18
		(i) Prove that $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - 3z)\hat{j} + (4x + cy + 2z)\hat{k}$ is solenoidal and determine constants a, b & c if $\vec{F}$ is rotational	
		(ii) Show that $\vec{F} = (ye^{xy} \cos z)\hat{i} + xe^{xy} \cos z\hat{j} - e^{xy} \sin z\hat{k}$ is conservative. Find its scalar potential ϕ . Hence find the work done in moving a particle in this field of force $\vec{F}$ from the point (0,0,0) to the point (-1,2, π)	

(iii)	Find the directional derivatives of $\phi = e^{2y} \cos(xz)$ at the origine along the tangent for the curve given by $x = a \cos t$, $y = a \sin t$, $z = at$ at $t = \frac{\pi}{4}$	
(iv)	Prove that $\nabla \left(\frac{\vec{a} \cdot \vec{r}}{r^n} \right) = \frac{\vec{a}}{r^n} - \frac{n(\vec{a} \cdot \vec{r})\vec{r}}{r^{n+2}}$ where $\vec{a}$ is a constant vector.	
B	Attempt any ONE of the following. (Vector Differentiation and integration)	7
(i)	Use the divergence theorem to evaluate $\oiint_S [(x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}] \cdot d\vec{S}$ over the cube bounded by the planes $x=0$, $x=1$, $y=0$, $y=1$, $z=0$ and $z=1$	
(ii)	Verify Green's theorem in the plane for $\oint_C (x^2 - xy)dx + (x^2 - y^2)dy$ where C is the boundary of the region enclosed by the curves $y = x^2$ and $y = x$.	